

PS Seminar 7

1. Repartiția unei v. a.
2. Schema binomială
3. Funcția de repartiție
4. Media și varianța

variabilă aleatoare

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$$(\Omega, \mathcal{F}) \xrightarrow{x} (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$$

spațiu probabilități

Aplicații

ex 1

Dintr-o urnă se conține 50 bile albe și 100 negre
se extrag cu revenire 3 bile

Fie X o variabilă aleatoare = nr. bilelor albe
obținute în cele 3 extrageri

a) Repartiția v.a. X

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$$X: \begin{pmatrix} 0 & 1 & 2 & 3 \\ p_0 & p_1 & p_2 & p_3 \end{pmatrix} \quad \swarrow \text{valori posibile}$$

50 albe

100 negre

150 total

$$p = \frac{50}{150} = \frac{1}{3} \quad \swarrow \text{extragere cu revenire} \Rightarrow \text{punct$$

$\swarrow$ prob. de succes

$$p_k = C_n^k \cdot \left(\frac{1}{3}\right)^k \cdot \left(\frac{2}{3}\right)^{3-k}, \quad h = \overline{0, 3}$$

$\swarrow$ schema binomială

$\swarrow$ prob. de
succes $n =$ nr. de extrageri

$$p_0 = C_3^0 \cdot \left(\frac{1}{3}\right)^0 \cdot \left(\frac{2}{3}\right)^3 = \frac{8}{27}$$

$$p_1 = C_3^1 \cdot \left(\frac{1}{3}\right)^1 \cdot \left(\frac{2}{3}\right)^2 = 3 \cdot \frac{1}{3} \cdot \frac{4}{9} = \frac{12}{27}$$

$$p_2 = C_3^2 \cdot \left(\frac{1}{3}\right)^2 \cdot \left(\frac{2}{3}\right)^1 = \frac{3 \cdot 2}{2} \cdot \frac{1}{9} \cdot \frac{2}{3} = \frac{6}{27}$$

$$p_3 = C_3^3 \cdot \left(\frac{1}{3}\right)^3 \cdot \left(\frac{2}{3}\right)^0 = 1 \cdot \frac{1}{27} \cdot 1 = \frac{1}{27}$$

$$X: \begin{pmatrix} 0 & \overset{\frac{1}{2}}{1} & \overset{\frac{2}{3}}{1} & 2 & 3 \\ \frac{8}{27} & \frac{12}{27} & \frac{6}{27} & \frac{1}{27} \end{pmatrix}$$

b)

$$P(X=2) = \frac{6}{27}$$

$$P(X=\bar{x}) = 0$$

$$P\left(X \geq \frac{1}{2}\right) = \frac{12}{27} + \frac{6}{27} + \frac{1}{27} = \frac{19}{27} \quad (\text{sum in comp.})$$

$$P\left(X < \frac{2}{3}\right) \Leftrightarrow P(X \leq 1) = \frac{8}{27} + \frac{12}{27} = \frac{20}{27}$$

$$P\left(\overset{A}{\underbrace{X < 2}} \mid \overset{\text{conditional}}{\underbrace{B}{X > 0.3}}\right) = \frac{P((X < 2) \cap (X > 0.3))}{P(X > 0.3)}$$

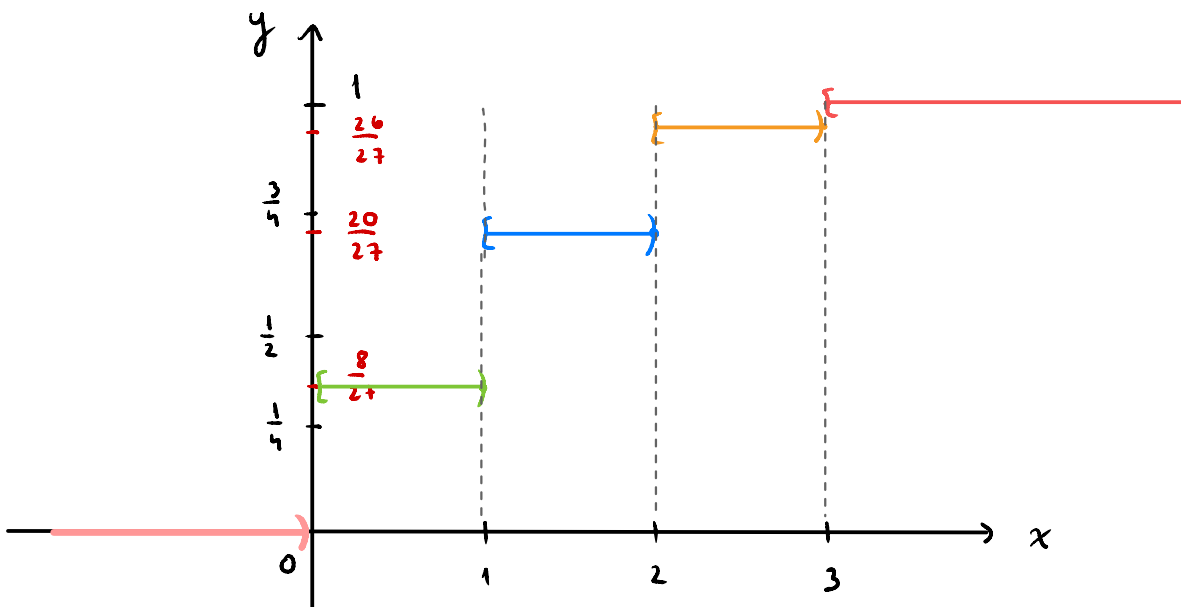
$$= \frac{P(0.3 < X < 2)}{1 - P(X < 0.3)} = \frac{\frac{12}{27}}{1 - \frac{8}{27}} = \frac{\frac{12}{27}}{\frac{19}{27}} = \frac{12}{19}$$

c) $F(x) = ?$ (funcția de repartiție)

$$X: \begin{pmatrix} \overset{x}{\text{red}} 0 & \overset{x}{\text{green}} 1 & \overset{x}{\text{blue}} 2 & \overset{x}{\text{orange}} 3 & \overset{x}{\text{red}} 4 \\ \frac{8}{27} & \frac{12}{27} & \frac{6}{27} & \frac{1}{27} & 0 \end{pmatrix}$$

$$F(x) = P(X \leq x) = \begin{cases} 0, & x < 0 \\ \frac{8}{27}, & 0 \leq x < 1 \\ \frac{8}{27} + \frac{12}{27}, & 1 \leq x < 2 \\ \left(\frac{8}{27} + \frac{12}{27}\right) + \frac{6}{27}, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

Reprezentare grafică



d) Calculati folosind functie de repartitie cum. prob.

$$\begin{aligned} & P(X < 2) \\ &= P((X \leq 2) \setminus \{X = 2\}) \\ &= P(X \leq 2) - P(X = 2) \\ &= F(2) - \frac{6}{27} = \frac{26}{27} - \frac{6}{27} = \frac{20}{27} \end{aligned}$$

cazul discret

$$P(a < X \leq b) = F(b) - F(a)$$

$$P(1 < X < 3) = F(3) - F(1) - P(X = 3)$$

$$P(1 \leq X < 2) = \underbrace{F(2) - F(1)}_{P(1 < X \leq 2)} - P(X = 2) + P(X = 1)$$

e) $E(X) = ?$ $Var(X) = ?$
 ? media ? varianta

$$\begin{aligned} E(X) &= 0 \cdot \frac{8}{27} + 1 \cdot \frac{12}{27} + 2 \cdot \frac{6}{27} + 3 \cdot \frac{1}{27} \\ &= \frac{1}{27} (12 + 12 + 3) = 1 \end{aligned}$$

$$\text{Var}(X) = E(X^2) - E(X)^2$$

$$X^2 = \begin{pmatrix} 0 & 1 & 4 & 9 \\ \frac{8}{27} & \frac{12}{27} & \frac{6}{27} & \frac{1}{27} \end{pmatrix}$$

$$\text{Var}(X) = \frac{1}{27} (12 + 4 \cdot 6 + 9 \cdot 1) - 1^2$$

$$= \frac{45}{27} - 1 = \frac{18}{27} = \frac{2}{3}$$

! $\text{Var}(X) \geq 0$ *menen*

$$\text{Deviation standard } \sigma = \sqrt{\text{Var}(X)} = \sqrt{\frac{2}{3}} = \frac{\sqrt{2} \cdot \sqrt{3}}{3}$$

Abbreve medie patrativă